

# SAM-RFI : An image segmentation approach to Radio Frequency Interference identification and flagging

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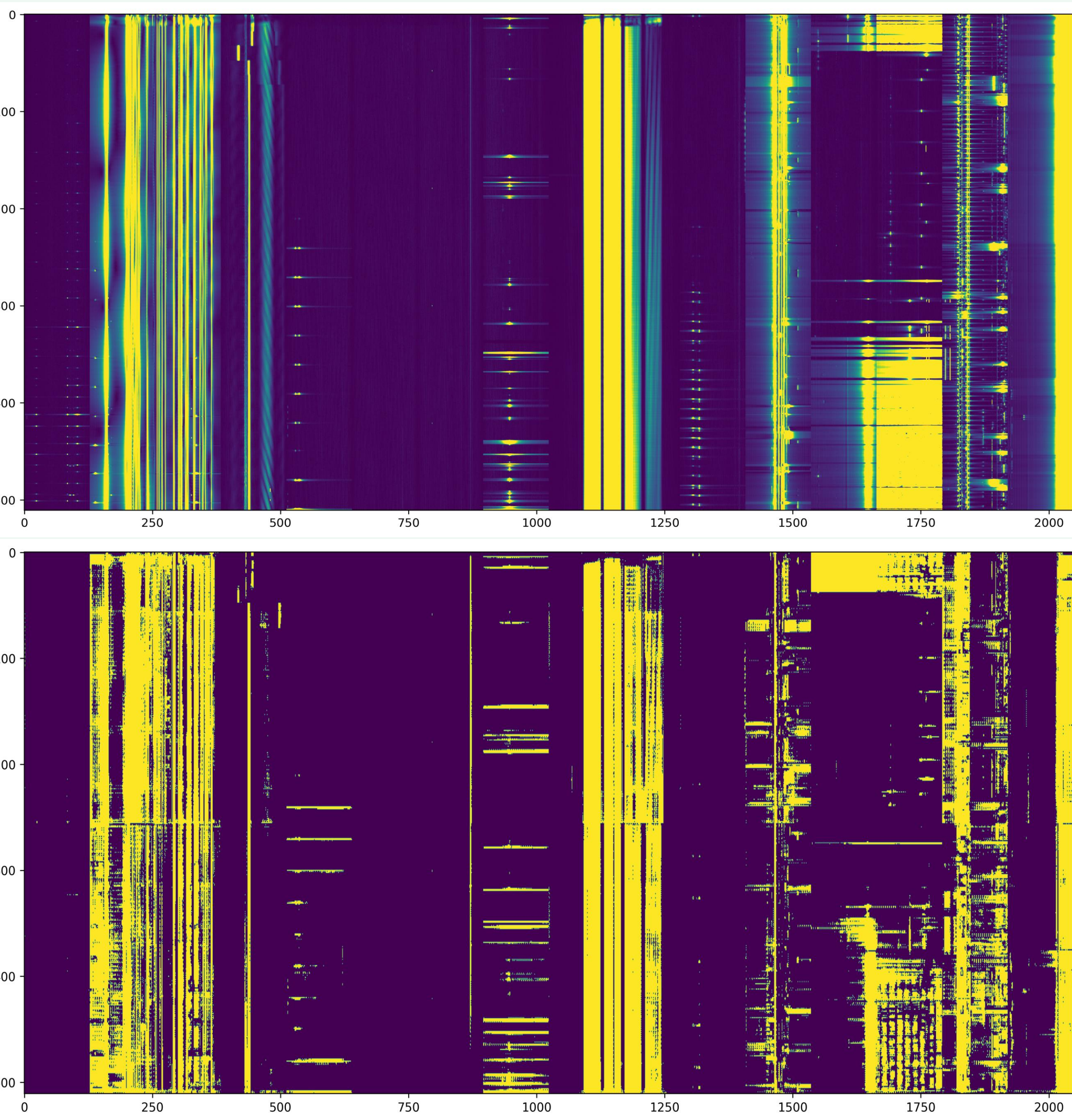


## Quick Takeaways

**The Problem:** Radio Frequency Interference (RFI) contaminates data, hindering accurate measurements. It is only getting worse with more human occupancy of the Radio Spectrum. We are looking for faint radio signals while looking up a booming bright RFI.

**Our Solution:** SAM-RFI leverages Meta's Segment Anything Model (SAM2) to intelligently flag and mitigate RFI. Provides tools for loading measurement sets, generating insightful waterfall plots, and even retraining custom SAM-RFI models.

**Semantic Segmentation:** Quantifiable metrics to demonstrate the effectiveness of our RFI flagging. Improvements in speed with lighter weight models with no loss in performance. [Resulting in cleaner data, improved accuracy... potentially faster](#)



Shown in the top panel of the image is the amplitude of the complex visibility data from the Very Large Array at 230-470 MHz (P-Band) of a calibrator source. The bright bands represent Radio Frequency Interference which saturates the receivers. The goal here is to accurately identify them on a 2D plane of time (x-axis) vs frequency sampled across 2k points (y-axis). The color scale spans 6 orders of magnitude and is plotted in log scale to show saturation. The primary source of interference shown here is satellite RFI. The bottom panel shows the segmentation output from the SAM-RFI software. It identifies a large fraction of the RFI outperforming thresholding-based flagging methods. This requires no human intervention or parameter tuning and results in flagged data which serves as an effective starting point for data reduction pipelines. Typically, data reduction pipelines spend about 20% of the time in flagging bad data.

## Abstract

Radio Frequency Interference (RFI), radio signals from electronics, permeates through astronomical observations using radio telescopes. The interference results in lower sensitivity and higher noise when present. Existing methods, such as TFCrop and RFlag, flag RFI to prevent the contamination of observations. However, these approaches are partially automatic and require manual input to flag RFI. We explore AI based image segmentation using Meta's Segment Anything Model (SAM) and its parameter space in segmenting RFI. We have developed an open-source pipeline called SamRFI, which segments and flags RFI from measurement sets (standard radio astronomy data format). We retrained SAM's weights from Hugging Face to fine-tune the model using Very Large Array data and synthetic injected with RFI. We define a metric, to measure the performance of our models based on sensitivity and overflagging. Retrained SAM models identify RFI structure and allow for further exploration in RFI segmentation.

## Training

The SamRFI pipeline leverages Meta's Segment Anything Model (SAM) for RFI segmentation. We retrained SAM using PyTorch and Hugging Face's transformers architecture to enhance its RFI detection capabilities. The retrained models utilized the phase calibrator 3C 147 from the P-Band data set (observed on February 17, 2015) and synthetic data with Gaussian persistent and intermittent RFI, combined with random noise to simulate sky background. To prepare the data, we loaded each baseline per polarization from the P-band measurement set, converted it into a time-frequency spectrogram, and created  $256 \times 256$  patches. We applied either LOG10 or SQRT stretching and normalized each patch by its median. Ground truth masks were generated using a threshold of 5 median absolute

deviations (MAD). We trained four models: SQRT Sigma 5, LOG10 Sigma 5, and a synthetic model, with sigma  $\sigma$  representing the number of median standard deviations used in creating the ground truth masks. Both SQRT Sigma 5 and LOG10 Sigma 5 were trained on 600 randomly selected patches with  $MAD \times 5$  ground truth masks.

## Validation and Testing

For validation, we used the 3C 129 P-Band data set. Each baseline was processed through each model, and a custom metric was used to evaluate flagging performance. We considered only XX or YY polarizations to compare baseline flagging with similar power and parallel hands. The spectrogram was sliced into  $256 \times 256$  patches, and each model generated a probability matrix indicating the confidence of detected RFI. Flags were created by applying a probability cutoff threshold.

## The Metric

To assess model performance on real data, we defined a comprehensive metric that evaluates sensitivity, overflagging, mean shifts, and standard deviation shifts. The metric calculation involves:

- Maxdev: maximum difference between the real data and the mean of leftover pixels
  - Fdiff: mean difference between flagged and leftover pixels
  - Sdiff: difference in median absolute deviations between leftover and flagged pixels
- The overall metric is the root sum square of four components: Sensitivity, Mean shifts, Median absolute deviation shifts, and Over-flagging (which is zero unless over-flagging exceeds 70%).

## More to explore

We plan to compare the performance of SamRFI with other deep learning-based flagging approaches, such as custom-trained CNNs. This will help us better understand the strengths and weaknesses of our model and identify areas for improvement. In our application, higher recall may be beneficial, even if it comes at the cost of precision, as it allows for more comprehensive RFI flagging.

Future work will focus on developing approaches for signal identification and modeling, with the goal of separating the source signal from contaminant RFI. This could involve exploring techniques such as signal decomposition, source separation algorithms, or more sophisticated modeling of RFI structures. By better understanding and characterizing the signals, we can improve the accuracy and effectiveness of RFI flagging.

We will explore alternative methods for reducing the dynamic range of our data, potentially improving the robustness and efficiency of our RFI flagging pipeline. This might involve investigating novel data transformations, normalization techniques, or other approaches that can help mitigate the impact of extreme values on our models.

We are always open to suggestions and collaborations. [Please don't hesitate to reach out to us.](#)